Elizabeth Kushelevsky

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Education

Barnard College, Columbia University – BA, Computer Science Expected graduation: December 2026

- 3.77 GPA; Minor in Russian; emphasis on software security and systems
- Coursework: Malware Analysis & Reverse Engineering, Operating Systems, Machine Learning Security, Applied Cryptography

Experience

Computer Science Teaching Assistant, Columbia University – New York, NY August 2025 - Present

- Teaching assistant for Advanced Programming (fall 2025, head TA fall 2026) and Advanced Systems Programming (spring 2026), both classes on Linux systems taught in C
- Explain complex concepts clearly in review sessions and office hours; grade student labs, homeworks, and exams
- Enhance grading infrastructure and homework resources; develop and test exam questions

Security Intern, The Health Co-Lab – Boston, MA May 2026 – August 2026

- Build a Rust/TypeScript application to prevent transmission of PHI through copy-and-paste actions and redirect to approved destinations via a HIPAA-compliant MCP server, supporting Windows, MacOS, and browser destinations
- Complete preparations for a SOC 2 audit of business and software practices, including creation and implementation of policies and monitoring mechanisms
- Assist detection and resolution of security issues in TypeScript and SQL code for an RCM platform for last-mile healthcare services

Projects

CTF Writeups egkushelevsky.xyz/ctf/

- Collection of solved CTF challenges, primarily reverse engineering and binary exploit
- Used tools such as Ghidra, IDA, and gdb to analyze programs and find vulnerabilities
- Designed attacks and wrote Python exploit scripts based on these vulnerabilities

Analysis of heady.exe gist.github.com

- Analysis of a malicious x86 DLL with backdoor functionality and the EXE which runs it to install persistent configuration data
- Used IDA and x32dbg to examine the files' obfuscation, registry access, and networking abilities

Secure messaging protocol github.com/egkushelevsky/secure-messaging

- Custom Rust protocol implementing authenticated encryption, AES in CBC mode, and a Diffie-Hellman double ratchet algorithm
- Wrote a TCP server using the tokio crate, establishing a TCP server, handling requests through asynchronous tasks, and enabling I/O multiplexing for each client

Extracurriculars and Skills

- C/C++, Python, Java, Rust, Linux systems, Windows systems, x86, TCP/IP, IDA
- **Deputy Copy Editor**, Columbia Daily Spectator
- **Executive Board,** CU Cyber
- **Member,** Rewriting the Code